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### **Non-Thermal Plasma Desulfurization of Petroleum Products and Method Thereof**

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#### **Abstract**

This invention discloses methods and processes to separate sulfur hydrocarbons in petroleum refinery feedstocks via non-thermal corona-discharged air (known as cold plasma). The procedure comprises physical and chemical processes including single or multi-step oxidation of nonpolar sulfur hydrocarbons by ozone bubbling and optionally the simultaneous addition of an extremely small amount of hydrogen peroxide. This is followed by an aqueous liquid/non-aqueous liquid extraction of the oxidized compounds under conditions sufficient to extract sulfur compounds into the aqueous extractant. This process is followed by a regeneration unit for the recovery of liquid extractant material. Moreover, a cooling tower is employed to prevent exhausting the vapors of hydrocarbon feedstocks, as a form of volatile organic compounds, during the plasma bubbling process. The invention introduces a desulfurization technique that effectively separates sulfur hydrocarbons from petroleum feedstocks and fuels, offering a complementary solution to traditional hydrodesulfurization processes or serving as a standalone system.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application claims the benefit of U.S. provisional patent application Ser. No. 63/618,310 filed on Jan. 6, 2024, the content of which incorporated herein by reference in its entirety.

### FIELD OF THE INVENTION

[0002] The present disclosure relates to a method for removing sulfur from hydrocarbon feedstocks, including light and heavy oil refinery products, natural gas condensate (NGC), naphtha, diesel, mazut, and straight-run diesel (SRD) via ozone (O<sub>3</sub>) bubbling carried out by non-thermal (cold) plasma technology.

### BACKGROUND

[0003] Desulfurization is a fundamental step in oil and gas refining, facilitating sulfur removal from hydrocarbon streams. Sulfur in crude oil is in the form of aliphatic compounds such as mercaptans and aliphatic sulfides and disulfides, as well as aromatic compounds such as thiophene, benzothiophene, dibenzothiophene, and alkyl derivatives such as 4,6-dimethyl-dibenzothiophene. While the primary method, hydrodesulfurization (HDS), achieves sulfur removal efficiently, it requires a high pressure (from about 20 atm. to about 140 atm.) and high temperature (from about 260° C. to about 430° C.). Moreover, conventional methods such as HDS involve large amounts of hazardous materials, solvents, costly metallic catalysts (noble metals such as cobalt and molybdenum), and hydrogen. Although efforts have been made to enhance HDS through catalyst modifications, the cost factor remains a significant limitation, further compounded by environmental concerns. Corrosion caused by sulfur further escalates refinery costs (e.g., maintenance) due to HDS equipment depreciation.

[0004] Navigating the global landscape of fuel sulfur content regulations reveals a committed move toward environmentally considerate standards. The U.S. Environmental Protection Agency (EPA) mandates a Tier 3 gasoline sulfur standard with an average of 10 ppm and a maximum of 80 ppm for individual batches. Similarly, the U.S. and Canada enforce a 15-ppm sulfur cap for on-road ultra-low sulfur diesel (ULSD). In the European Union, the Euro 5 and Euro 6 standards restrict sulfur content to 10 ppm for gasoline and diesel. Concurrently, numerous Middle Eastern nations, including Saudi Arabia, U A E, and Kuwait, align with the EU's 10-ppm benchmark for both fuel types. Hence, it is paramount to develop efficient, cost-effective, and eco-friendly desulfurization techniques that can supplement or replace conventional methods like HDS to achieve the stringent 10-15 ppm sulfur limits consistently.

[0005] In the traditional HDS method, removing sulfur from hydrocarbons also presents a challenge from a chemistry perspective. The robust covalent bonds of sulfur and the spatial hindrance around these molecules on the catalyst surface complicate their conversion into hydrogen sulfide (H<sub>2</sub>S) within standard HDS processes. This is especially the case for dibenzothiophene and 4,6-dimethyl-dibenzothiophene. Overcoming this intrinsic barrier to traditional HDS methodologies necessitates advanced desulfurization techniques. Consequently, contemporary

research and development focus has shifted towards exploring physicochemical alternatives such as selective adsorptive mechanisms, extraction via ionic liquids, electrochemical redox processes, and low-temperature plasma-catalyzed desulfurization.

[0006] Several attempts have also been made to explore catalytic and oxidative desulfurization without using expensive hydrogen. Examples include U.S. Pat. Nos. 3,551,328A, 6,673,230B2, 8,920,635B2, US20040154959A1, and WO2013052102A1, each of which is herein incorporated by reference in its entirety. While the processes adopted in each of the above are seemingly effective, they pose economic and environmental challenges. Specifically, they often require substantial quantities of hazardous oxidants and catalysts and produce significant amounts of volatile organic compounds (VOCs). These methods have remained in the research or pilot stages. Moreover, some of these methods are suited for natural gas products and not applicable to oil feedstocks. Their broad adoption hinges on the efficiency and economic viability of the materials and technologies employed.

[0007] U.S. Pat. No. 9,365,780B2, the entire content of which is incorporated herein by reference, discloses a lab-scale process for removing sulfur compounds from a petroleum product of straight-run diesel (SRD) ( $\cong 1.41$  mass %) by using ozone ( $O_3$ ) as the oxidizer. The process involves the oxidation of sulfur species by ozone bubbling at room temperature. Since the process is on a lab scale, the evaporation of the feed is minimal, and most of the evaporated feed recirculates within the laboratory setup (such as in Erlenmeyer flasks or similar apparatuses). Therefore, the process may not be able to address a substantially higher evaporation volume occurring at an industrial scale. Further, the solvents used during the process are pure or ultra-pure ones, which do not adequately mimic the solvents used at an industrial scale. Finally, the process utilizes pure oxygen-ozone (not produced from the air) or a combination of oxygen-ozone with tert-butyl hydroperoxide t-BUOOH,  $KMnO_4$ , EDTA, or blends such as  $H_2O_2$ /acetic acid or formic acid, which if used at an industrial scale, would make the industrial process financially unfeasible.

[0008] In view of the above, there is a need for providing an industrial-scale process that is environmentally friendly, financially justifiable, and easy to implement for removing sulfur from a wide range of hydrocarbon streams as discussed above.

## SUMMARY

[0009] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key or essential aspects of the claimed subject matter.

[0010] Broadly stated, in some embodiments, the present disclosure relates to a process for separating sulfur compounds from hydrocarbon feedstock, the process comprising: [0011] providing the hydrocarbon feedstock; [0012] bubbling the hydrocarbon feedstock in a plasma oxidation reaction system with an ozone-rich stream, as a primary oxidizer, to obtain treated hydrocarbon, the treated hydrocarbon comprising oxidized sulfur compounds; and [0013] separating the oxidized sulfur compounds from the treated hydrocarbon to obtain a desulfurized product; [0014] wherein the process is free of added metal-based catalysts or hydrogen.

[0015] In some embodiments, the process includes one or more of the following features: [0016] the hydrocarbon feedstock is NGC, naphtha, diesel, mazut, kerosene, fuel oil, SRD or combinations thereof, [0017] wherein the sulfur compounds comprise aliphatic or refractory aromatic compounds such as mercaptans, thiophenes, benzo-thiophenes, and dibenzothiophenes [0018] further comprising adding a secondary oxidizing compound during the bubbling step [0019] the secondary oxidizing compound comprises at least one of hydrogen peroxide, sulfuric acid, potassium permanganate, acetic acid and nitric acid [0020] the bubbling step is non-thermal [0021] the ozone-rich stream is corona discharged [0022] the separating step comprises liquid-liquid extraction [0023] bubbles in contact with the hydrocarbon feedstock, in the bubbling step, are in the form of micro-bubbles [0024] the micro-bubbles have a size range from about 10 to about 100 micrometers [0025] the process is carried out in continuous form or non-continuous form [0026] the plasma

oxidation reaction system comprises one or more plasma oxidation reactors [0027] the bubbling in each of the one or more plasma oxidation reactors is carried out at a predetermined height of hydrocarbon feedstock, the predetermined height ranging from about 1.5 m to about 4.5 m [0028] further comprising returning vapors of hydrocarbon components during the bubbling steps back to the plasma oxidation reactor [0029] the separating step comprise contacting the oxidized sulfur compounds with a polar solvent, wherein the polar solvent comprises a caustic solution, methanol, dimethyl formamide, or acetonitrile [0030] the separating step is carried out at a duration from about 30 min to about 2 hours [0031] a desulfurization efficiency of up to 90% is achieved [0032] Broadly stated, in some embodiments, the present disclosure relates to a system configured for separating sulfur compounds from a hydrocarbon feedstock, the system comprising: [0033] a reaction vessel for receiving the hydrocarbon feed stock, the reaction vessel comprising an inlet for the hydrocarbon feedstock, an outlet for treated hydrocarbon; [0034] at least one nozzle at or proximate to the bottom of the reaction vessel configured to deliver an ozone-rich stream, as a primary oxidizer and in the form of bubbles, to the reaction vessel containing the hydrocarbon feedstock; and [0035] at least one extraction unit for separating sulfur compounds from the treated hydrocarbon exiting the reaction vessel.

[0036] In some embodiments, the system includes one or more of the following features: [0037] the reaction vessel operates at a predetermined height of the hydrocarbon feedstock therein, the predetermined height ranging from about 1.5 m to about 4.5 m. [0038] the bubbles are micro-bubbles having a size range from 10 micrometers to 100 micrometers

[0039] Broadly stated, in some embodiments, the present disclosure relates to a desulfurization module comprising: [0040] a hydrocarbon feedstock reservoir; and [0041] a bubbling means for delivering an ozone-rich stream, as a primary oxidizer and in the form of bubbles, to the hydrocarbon feedstock reservoir, [0042] wherein the module is free of added metal-based catalysts, water or hydrogen.

[0043] In some embodiments, the desulfurization module may operate upstream of a secondary desulfurization plant.

[0044] All features of exemplary embodiments which are described in this disclosure and are not mutually exclusive can be combined with one another. Elements of one embodiment can be utilized in the other embodiments without further mention. Other aspects and features of the present invention will become apparent to those ordinarily skilled in the art upon review of the following description of specific embodiments in conjunction with the accompanying Figures.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0045] A detailed description of specific exemplary embodiments is provided herein below with reference to the accompanying drawings in which:

[0046] FIG. 1 is a non-limiting example of a plasma-assisted oxidative desulfurization process according to an embodiment of the present invention;

[0047] FIG. 2 is a non-limiting example of a plasma-assisted oxidative desulfurization process according to another embodiment of the present invention;

[0048] FIG. 3 is a non-limiting flowchart of the plasma-based desulfurization steps in accordance with an embodiment of the present disclosure.

[0049] In the drawings, exemplary embodiments are illustrated by way of example. It is to be expressly understood that the description and drawings are only for the purpose of illustrating certain embodiments and are an aid for understanding. They are not intended to be a definition of the limits of the invention.

### DETAILED DESCRIPTION

[0050] The present technology is explained in greater detail below. This description is not intended to be a detailed catalog of all the different ways in which the technology may be implemented or all the features that may be added to the instant technology. For example, features illustrated with respect to one embodiment may be incorporated into other embodiments, and features illustrated with respect to a particular embodiment may be deleted from that embodiment. In addition, numerous variations and additions to the various embodiments suggested herein will be apparent to those skilled in the art, considering the instant disclosure, which variations and additions do not depart from the present technology. Hence, the following description is intended to illustrate some embodiments of the technology and not to specify all permutations, combinations, and variations thereof exhaustively.

[0051] The inventors in the present invention have developed a desulfurization process that utilizes plasma technology to desulfurize light and heavy oil refinery products in an economically viable and environmentally friendly manner. Advantageously: [0052] the process according to the present invention eliminates the use of costly catalysts and their disposals which is an environmental concern as well; [0053] the process, according to the present invention, eliminates the need for hydrogen, which is supplied by a costly hydrogen production unit (HPU) in traditional desulfurization units. It is known that use of common HPUs results in GHG emissions which is also a major environmental concern whereas the process according to the present invention uses electricity (for plasma generation) and therefore, GHG production is substantially limited; [0054] the process according to the present invention results in rapid reaction times at ambient temperature and atmospheric pressure as opposed to high-pressure and high-temperature traditional processes. More advantageously, the ozonation (bubbling) process, according to the present invention, has the ability to oxidize aliphatic compounds such as aliphatic sulfides, disulfides, and mercaptans as well as aromatic compounds such as thiophene, benzothiophene, dibenzothiophene, and alkyls such as 4,6-dimethyl-dibenzothiophene. The oxidation, according to the present invention, is achieved using an ozone-rich stream, as a primary oxidizer, that is bubbled through a hydrocarbon feedstock and particularly through light and heavy oil refinery products; and [0055] The present invention introduces a desulfurization technique that effectively separates sulfur hydrocarbons from petroleum feedstocks and fuels, offering a complementary solution to traditional HDS processes or serving as a standalone system.

[0056] The hydrocarbon feedstock may include, but not limited to NGC, naphtha, diesel, kerosene, mazut, fuel oil, SRD or combinations thereof.

[0057] In some embodiments, the ozone-rich stream is corona-discharged air, rich in ozone (O.sub.3) and other reactive oxidants

[0058] In some embodiments, the oxidation by ozone bubbling is followed by the addition of a secondary oxidizing compound converting thiophene compounds into polar soluble substances.

[0059] Examples of the secondary oxidizing compound include but are not limited to hydrogen peroxide, sulfuric acid, potassium permanganate, acetic acid and nitric acid.

[0060] In a preferred embodiment, the secondary oxidizing compound is hydrogen peroxide (H.sub.2O.sub.2).

[0061] In another alternative embodiment, no secondary oxidizing compound is used.

[0062] Advantageously, in another embodiment, the consumption of hydrogen peroxide in the process, according to the present invention, is substantially less than other comparative processes.

[0063] In the desulfurization process according to the present invention, the size of ozone microbubbles (10-100 micrometers) is crucial in enhancing the efficiency of ozone transfer in hydrocarbon feedstocks due to their larger surface area per unit volume. This attribute accelerates mass transfer and dissolution, leading to faster reaction kinetics, particularly beneficial in the oxidation of sulfur compounds. Microbubbles also ensure a more uniform distribution of ozone throughout the hydrocarbon feed, promoting consistent treatment. Additionally, these microbubbles require less energy to generate, potentially offering a more energy-efficient solution.

[0064] When ozone is dissolved in a liquid, it closely adheres to Henry's Law. Consequently, determining a saturation ratio (dissolved gas-to-liquid volume ratio) is essential. The height of ozone bubbling depends on various factors, including the ozone delivery system's design, ozone flow rate, and bubble size. Considerations encompass the design of the ozone delivery system, bubble size, flow rate, and temperature and pressure. Specifically, smaller bubbles tend to ascend more slowly and disperse over a reduced vertical distance in comparison to larger bubbles. Additionally, temperature and pressure can impact the solubility of gases, ozone included, and variations in these parameters may influence the behavior of microbubbles in the feed. In the designed system, the temperature and pressure are under ambient conditions, the ozone delivery system utilizes titanium diffusers for creating microbubbles, the flow rate of the feedstock for ozonation process ranges between 100 and 600 liter per minute (L/min), and the height of ozone bubbling column is set between 1.5 to 4.5 meters.

[0065] Once the sulfur is oxidized as described above, the sulfur compounds become more soluble in polar aqueous solvents, such as caustic soda solvent, dimethyl formamide, or acetonitrile. As a result, these sulfur compounds can be efficiently separated in a mixer tank using traditional liquid/liquid extraction techniques, including methods like the Merox process, given appropriate conditions and durations. The efficacy of sulfur extraction, achieved by bringing the oxidized feed into contact with aqueous liquid extractors, hinges on the solubility of the oxidized sulfur compounds in the chosen solvent. Consequently, the timing and conditions of the oxidation step are of paramount importance.

[0066] FIG. 1 is a non-limiting example of a plasma-assisted oxidative desulfurization process **100** according to the present invention (which may alternatively be known as a treatment sequence) and based on the concept described above.

[0067] First a hydrocarbon feedstock **110** comprising sulfur compounds is introduced into a first oxidative reactor **115** wherein the hydrocarbon feedstock is subjected to ozone, as a primary oxidizer, being bubbled into the hydrocarbon feedstock in a non-thermal manner (no heating). The ozone is from an ozone-rich stream **125** originating from an ozone source **120**. In one embodiment, the ozone source **120** is a plasma corona discharge reactor ozone source.

[0068] In some embodiment, a secondary oxidizing compound, such as  $\text{H}_2\text{O}_2$ , is introduced into the first oxidative reactor **115** to be added to the hydrocarbon feedstock **110** ensuring that sulfur components are effectively extracted at ambient temperature and pressure.

[0069] The mixing between  $\text{O}_3$  and  $\text{H}_2\text{O}_2$  leads to a very high  $\text{O}_3$  decomposition rate to  $\text{OH}^\bullet$  radicals, concomitant with a high hydroxyl ( $\text{OH}^\bullet$ ) radical exposure.  $\text{O}_3$  and  $\text{H}_2\text{O}_2$  can generate hydroxyl radicals that are strong oxidizing agents. Ozone and the hydroxyl radical are oxidants, but the oxidative property of the hydroxyl radical is more than that of  $\text{O}_3$ .

[0070] The bubbling that occurs in the first oxidative reactor **115** causes a fraction of the hydrocarbon feedstock **110** to escape the reactor as vapor or vent gas (V) while a cooling tower (CT) returns the vapor or vent gas (V) into the first oxidative reactor **115**.

[0071] Next, the outlet stream **130** from the first oxidative reactor that contains oxidized sulfur compounds (due to ozone bubbling) flows into a liquid-liquid extraction unit **135** in which the stream **130** is mixed with one or more extracting solvents **132** selected from the group consisting of dimethylformamide, caustic solutions, methanol and acetonitrile for a given period of time to so that the liquid-liquid extraction process results in two distinct liquid phases. One liquid phase is the sulfur depleted hydrocarbon stream **140** that may be the final desulfurized product, and the other liquid phase is a sulfur-rich stream **145** that comprises extracted sulfur compounds and the solvent. The sulfur-rich stream **145** may be further introduced to a regeneration unit **150** to process the sulfur-rich stream **145** and recover solvent **132** for recycle into the liquid-liquid extraction unit **135** and separate the sulfur compounds **155**.

[0072] The regeneration unit **150** enables recycling and reuse of solvent to minimize environmental

concerns. Prior to solvent recycling, some precipitates in the sulfur-rich stream **145** are removed by filtration within the regeneration unit. The regenerated solvent is then evaporated using rotary evaporation, and the remaining residue is processed with organic solvents. The regenerated solvent may then be recycled without substantial activity loss through these procedures.

[0073] The sulfur depleted hydrocarbon stream **140** may contain sulfur compounds at a concentration that is compliant with regulatory requirements or other desirable specifications. This can be measured by a sulfur analyzer(S) shown in FIG. **1** that analyzes sulfur content in the sulfur depleted hydrocarbon stream **140**. In this case, wherein the sulfur concentration is compliant with regulatory requirements or other desirable specifications, the desulfurization process as shown in non-limiting embodiment of FIG. **1** is deemed as complete.

[0074] In some embodiments, if the sulfur concentration in the sulfur depleted hydrocarbon stream **140**, resulting from the first treatment sequence, is still above the regulatory requirements or other desirable specifications, then the sulfur depleted hydrocarbon stream **140** would be subjected to a distillation unit (not shown) which will separate more sulfur compounds until the acceptable threshold of sulfur content in the sulfur depleted hydrocarbon stream **140** is reached.

[0075] In another embodiment, if the sulfur concentration in the sulfur depleted hydrocarbon stream **140**, resulting from the first treatment sequence, is still above the regulatory requirements or other desirable specifications, then the sulfur depleted hydrocarbon stream **140** would be considered as feed for another desulfurization process (i.e. a second treatment sequence) that is identical or substantially identical to the process of FIG. **1**, i.e. comprising an oxidative reactor in which ozone bubbling takes places and the reactor outlet would be sent to a liquid-liquid extraction unit using solvents as described above (see Example 4). Repeated desulfurization modules would ensure that a sulfur depleted hydrocarbon stream from the last desulfurization unit comprises sulfur concentrations that is compliant with regulatory requirements or other desirable specifications.

[0076] In yet another embodiment, if the sulfur concentration in the sulfur depleted hydrocarbon stream **140**, resulting from the first treatment sequence, is still above the regulatory requirements or other desirable specifications, then the sulfur depleted hydrocarbon stream **140** would be subjected to another liquid-liquid extraction unit. This is shown in FIG. **2** in which one single oxidative reactor **115** (for ozone bubbling) is followed by two consecutive liquid-liquid extraction units **135A-B** until the sulfur concentration in the sulfur depleted hydrocarbon stream **142** matches the regulatory requirements or other desirable specifications. In this embodiment, the first liquid-liquid extraction unit may use fresh solvent **132** and the second liquid-liquid extraction unit may use a regenerated solvent from regeneration unit **150**. Alternatively, both the first and the second liquid-liquid extraction units may use fresh solvent as well as regenerated solvent in combination.

[0077] The assessment as to whether a second desulfurization unit (i.e. a second treatment sequence) is also shown by the flowchart in FIG. **3**. As shown in FIG. **3**, for a high sulfur hydrocarbon feed that is subjected to a first desulfurization unit (for example the desulfurization process as shown in FIG. **1**), if the sulfur content in the desulfurized stream from the first desulfurization unit is within an acceptable range, the desulfurized stream from the first desulfurization unit can be deemed as the final desulfurized product. However, if the sulfur content in the desulfurized stream from the first desulfurization unit is not within an acceptable range, more desulfurization units (which may only include an additional liquid-liquid extraction unit with fresh solvent) are needed to bring down the sulfur content to an acceptable range.

[0078] It would be apparent to the person skilled in the art that other variations in the process such as multiple ozone bubbling steps each followed by one or more liquid-liquid-extraction steps could still be contemplated without departing from the spirit of the present invention.

[0079] The description above will be further elaborated in the following examples.

## EXAMPLES

[0080] The feedstocks tested in the following examples, were obtained from petroleum refineries. All the samples were maintained under ambient temperature (20° C.-25° C.) and atmospheric

pressure before the experiments. The ozone (O.sub.3) generator for non-thermal plasma discharge was designed and manufactured in house. In the laboratory-scale tests, the airflow rate was varied between 5 and 8 liters per minute (L/min), with the ozone concentration adjusted based on the efficiency of the ozone generator. In the examples below, the NGC, mazut, and SRD samples were ozone-bubbled with and without H.sub.2O.sub.2 at ambient temperature and atmospheric pressure for different periods, followed by sulfur extraction using different polar solvents. Sulfur extraction was done using industrial grade dimethylformamide (DMF) and caustic soda (NaOH).

Example 1

[0081] In this Example, a single-step ozone bubbling for an NGC sample was conducted in a sealed 2-liter glass bottle with a 5-8 L/min flow rate. Below are the results for the NGC sample after plasma treatment and caustic soda extraction steps with and without a distillation process: The sample volume was 1.5 L, with a sample-to-caustic soda volume ratio of 1:1 v/v. In this specific example, H.sub.2O.sub.2 (as a secondary oxidizing compound) was not added during the ozone bubbling process. The Tanaka Scientific RX-360SH sulfur meter was employed to assess the total sulfur content before and after desulfurization. The initial sulfur concentration was measured at 1572 ppm.

[0082] Table 1 illustrates that the desulfurization efficiency of the NGC sample is 19% before distillation and 51.5% after distillation. These results reveal that for the extraction with caustic soda and bubbling, even without the use of H.sub.2O.sub.2, some oxidized sulfur compounds remain dissolved in the NGC sample without distillation. These compounds are effectively separated from NGC only when extraction is followed by distillation.

TABLE-US-00001

| TABLE 1 Total sulfur analysis results for NGC in lab-scale experiments with and without distillation (No added H.sub.2O.sub.2 as secondary oxidizer). Desulfurization efficiency          |   |   |      |    |     |      |  |  |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|------|----|-----|------|--|--|--|
| Initial bubbling No. of Without (without With efficiency (with sulfur time bubbling distillation distillation) distillation distillation) Sample [ppm] [hr] sequences [ppm] [%] [ppm] [%] |   |   |      |    |     |      |  |  |  |
| NGC 1572                                                                                                                                                                                  | 1 | 1 | 1267 | 19 | 761 | 51.5 |  |  |  |

Example 2

[0083] In this Example, a single-step ozone bubbling was conducted for 30 minutes in 1.5 L of an NGC sample at a 5-8 L/min flow rate. In this example, 0.1 wt. % of H.sub.2O.sub.2 was added during the ozone bubbling process. Furthermore, instead of caustic soda, the polar sulfur compounds in the ozonized NGC sample were extracted using DMF (1:1 v/v) as solvent. The bubbling time also decreased by half (30 min) as compared to Example 1. As shown in Table 2, the measurements indicate a desulfurization efficiency improvement to 75%, representing a fourfold enhancement in this scenario. This example demonstrates the impact of using H.sub.2O.sub.2 in conjunction with oxidation and extraction methods on enhancing desulfurization efficiency.

TABLE-US-00002

| TABLE 2 Total sulfur analysis results for NGC in lab- scale experiments without distillation. Total After Initial bubbling No. of desulfuriza- Desulfurization sulfur time bubbling tion efficiency Sample [ppm] [hr] sequences [ppm] [%] |     |   |     |    |  |  |  |  |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|-----|----|--|--|--|--|--|
| NGC 1572                                                                                                                                                                                                                                  | 0.5 | 1 | 389 | 75 |  |  |  |  |  |

Example 3

[0084] This Example presents findings from a single-step plasma treatment of an NGC sample following DMF extraction but with a longer ozone bubbling time. The sample volume size was 1.0 L and the NGC/DMF ratio was 1:1 v/v. 0.1 wt. % of H.sub.2O.sub.2 was added during the ozone bubbling process. Table 3 illustrates that extending the bubbling duration increases the total sulfur removal efficiency to 78%.

TABLE-US-00003

| TABLE 3 Total sulfur analysis results for NGC in lab-scale without distillation. Total After Initial bubbling No. of desulfuriza- Desulfurization sulfur time bubbling tion efficiency Sample [ppm] [hr] sequences [ppm] [%] |     |   |     |    |  |  |  |  |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|-----|----|--|--|--|--|--|
| NGC 1572                                                                                                                                                                                                                     | 1.3 | 1 | 345 | 78 |  |  |  |  |  |

Example 4

[0085] In this Example, ozone bubbling was conducted in two sequences each lasting for 120 min in 1.5 L of a high-sulfur SRD sample at a 5-8 L/min flow rate. Similarly, 0.1 wt. % of



H.sub.2O.sub.2 was added, as secondary oxidizing compound, during the ozone bubbling process, and the polar sulfur compounds in the ozonized SRD sample were extracted using DMF (1:1 v/v). Table 4 presents the results obtained from two treatments of SRD samples, conducted without any distillation process. For the SRD sample with an initial sulfur content of 11282 ppm, the desulfurization efficiency reached 36% after the first treatment and increased to 54% following the second treatment.

TABLE-US-00004 TABLE 4 Total sulfur analysis results for SRD in lab-scale without distillation. Total Initial bubbling No. of 1.sup.st Desulfurization 2.sup.nd Desulfurization sulfur time bubbling Sequence efficiency Sequence efficiency Sample [ppm] [hr] sequences [ppm] [%] [ppm] [%] SRD 11282 4 2 7193 36 5182 54

Example 5

[0086] In this Example, ozone bubbling was conducted in one step for a total period of 8 hours in 1.5 liter of a high-sulfur mazut sample at a 5-8 L/min flow rate. No H.sub.2O.sub.2 was added. The polar sulfur compounds in the ozonized mazut sample were extracted using DMF (0.1:1 v/v). Table 5 presents the results obtained for this sample conducted without any distillation process. For the mazut sample with an initial sulfur content of .sup.~35000 ppm, the desulfurization efficiency reached 57.1% after the extraction.

TABLE-US-00005 TABLE 5 Total sulfur analysis results for mazut in lab-scale (No H.sub.2O.sub.2 or distillation). Total After Initial bubbling No. of desulfuriza- Desulfurization sulfur time bubbling tion efficiency Sample [ppm] [hr] sequences [ppm] [%] mazut 35000 8 1 15000 57.1

Example 6

[0087] In this Example, ozone bubbling was conducted in two sequences for a total bubbling period of 14 hours in 1.5 L of a high-sulfur SRD sample at a 5-8 L/min flow rate. Each treatment sequence includes ozonation (in the oxidative reactor) followed by two consecutive liquid-liquid extraction units.

[0088] No H.sub.2O.sub.2 was added. The polar sulfur compounds in the ozonized SRD sample were extracted using DMF (0.5:1 v/v) for the two consecutive extractions after each ozone bubbling step. Table 6 presents the results obtained from two treatments of SRD samples conducted without any distillation process. For the SRD sample with an initial sulfur content of 19802 ppm, the desulfurization efficiency reached 61.5% after the first treatment sequence and increased to 89% following the second treatment sequence.

TABLE-US-00006 TABLE 6 Total sulfur analysis results for SRD in lab-scale (No H.sub.2O.sub.2 or distillation). Total Initial bubbling No. of 1.sup.st Desulfurization 2.sup.nd Desulfurization sulfur time bubbling step efficiency step efficiency Sample [ppm] [hr] sequences [ppm] [%] [ppm] [%] SRD 19802 14 2 7632 61.5 2179 89

Example 7

[0089] In this industry-scale Example, ozone bubbling was conducted in one step for a total period of 24 hours in 5 metric tons of high-sulfur gas oil at a 500 L/min flow rate. The desulfurization process was conducted in an industrial pilot-plant. After 24 hours of treatment, the polar sulfur compounds in the ozonized gas oil sample were extracted two times using DMF (0.1:1 v/v). Table 7 presents the results without adding any H.sub.2O.sub.2 during the ozonation process or distillation step. For the gas oil with an initial sulfur content of 11890 ppm, the desulfurization efficiency reached 75.7% after the first extraction and increased to 87.7% following the second extraction.

TABLE-US-00007 TABLE 7 Total sulfur analysis results for gas oil in industrial scale (No H.sub.2O.sub.2 or distillation). Total Initial bubbling No. of After 1.sup.st Desulfurization After 2.sup.nd Desulfurization sulfur time bubbling extraction efficiency extraction efficiency Sample [ppm] [hr] sequences [ppm] [%] [ppm] [%] Gas oil 1189 24 2 2886 75.7 1457 87.7

[0090] While illustrative and/or presently preferred embodiments of the invention have been

described in detail hereinabove, it is to be understood that the inventive concepts may be otherwise variously embodied and employed and that the appended claims are intended to be construed to include such variations except insofar as limited by the prior art.

## Claims

1. A process for separating sulfur compounds from hydrocarbon feedstock thereby obtaining a sulfur-depleted product, the process comprising: providing the hydrocarbon feedstock; bubbling the hydrocarbon feedstock in a plasma oxidation reaction system with an ozone-rich stream, as a primary oxidizer, to obtain treated hydrocarbon, the treated hydrocarbon comprising oxidized sulfur compounds; and separating the oxidized sulfur compounds from the treated hydrocarbon to obtain the sulfur-depleted product; wherein the process is free of added metal-based catalysts or hydrogen.
2. The process according to claim 1, wherein the hydrocarbon feedstock is NGC, naphtha, diesel, kerosene, fuel oil, SRD or combinations thereof.
3. The process according to claim 1, wherein the sulfur compounds comprise aliphatic or refractory aromatic compounds such as mercaptans, thiophenes, benzo-thiophenes, and dibenzothiophenes.
4. The process according to claim 1, further comprising: adding a secondary oxidizing compound during the bubbling step comprising at least one of hydrogen peroxide, sulfuric acid, potassium permanganate and nitric acid.
5. The process according to claim 1, wherein the bubbling step is non-thermal.
6. The process according to claim 1, wherein the ozone-rich stream is corona discharged.
7. The process according to claim 1, wherein the separating step comprises liquid-liquid extraction.
8. The process according to claim 1, wherein bubbles in contact with the hydrocarbon feedstock, in the bubbling step, are in the form of micro-bubbles having a size range from 10 to 100 micrometers.
9. The process according to claim 1, wherein the process is carried out in continuous form or non-continuous form.
10. The process according to claim 1, wherein the plasma oxidation reaction system comprises one or more plasma oxidation reactors.
11. The process according to claim 1, wherein the bubbling in each of the one or more plasma oxidation reactors is carried out at a predetermined height of hydrocarbon feedstock, the predetermined height ranging from about 1.5 m to about 4.5 m.
12. The process according to claim 1, further comprising: returning vapors of hydrocarbon components during the bubbling steps back to the plasma oxidation reactor.
13. The process according to claim 1, wherein the separating step comprises contacting the oxidized sulfur compounds with a polar solvent, wherein the polar solvent comprises a caustic solution, methanol, dimethyl formamide or acetonitrile.
14. The process according to claim 1, wherein the separating step is carried out at a duration from about 30 min to about 2 hours.
15. The process according to claim 1, wherein a desulfurization efficiency of up to 90% is achieved.
16. A system configured for separating sulfur compounds from a hydrocarbon feedstock, the system comprising: a reaction vessel for receiving the hydrocarbon feed stock, the reaction vessel comprising an inlet for the hydrocarbon feedstock, an outlet for treated hydrocarbon; at least one nozzle at or proximate to the bottom of the reaction vessel configured to deliver an ozone-rich stream, in the form of bubbles, to the reaction vessel containing the hydrocarbon feedstock; and an extraction unit for separating sulfur compounds from the treated hydrocarbon exiting the reaction vessel.
17. The system according to claim 16, wherein the reaction vessel operates at a predetermined

height of the hydrocarbon feedstock therein, the predetermined height ranging from about 1.5 m to about 4.5 m.

**18.** The system according to claim 16, wherein the bubbles are micro-bubbles having a size range from about 10 micrometers to about 100 micrometers.

**19.** The system according to claim 16, wherein the system operates upstream or downstream of a traditional desulfurization plant.

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